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Design of a Robot for carrying out research on hybrid robot's mobility: case of a mecanum wheel-legged robot

Kouame Yann Olivier A.
School of Electronics and Communication engineering, REVA University
Bangalore, INDIA
akansiekouame@gmail.com

Rajashekhar C. Biradar
School of Electronics and Communication engineering, REVA University
Bangalore, INDIA
rcbiradar@reva.edu.in

R. Karthik
School of Electronics and Communication engineering, REVA University
Bangalore, INDIA
Karthik.rajendra@reva.edu.in

Geeta D. Devanagavi
School of Computing and Information Technology, REVA University
Bangalore, INDIA
dgeetha@reva.edu.in

***Abstract*— The advancement of technology favors the creation of intelligent machines capable of executing complex tasks. Automated machines that can be teleoperated or autonomous open up a vast pool of possibilities, among which substituting humans in some instances where their safety is endangered, which is one of the significant advantages. Having the ability to substitute humans sets some requirements like object manipulation, mobility, and intelligence. Various mechanisms are employed to provide mobility to the machines considering the mobility aspect. The most common mobile robots are quadruped robots because of their advantages. Over the years, many ideas have been proposed for quadruped robot designs to address specific problems. An improvement of quadruped robots is hybrid wheel-legged robots, on which researches are still ongoing—this paper attempt to address the robots' mobility problem with a mecanum wheel-legged robot design proposal. The paper discusses the design concerning three aspects: mechanical, electronic, and software. The targeted goal is to provide a robot platform that can be used to conduct research on wheel-legged robot mobility.**

Keywords—mecanum wheel robot, quadruped robot, wheel-legged robot, hybrid locomotion robot, hybrid robot's mobility

I. INTRODUCTION

A *robot* can be defined as a computer-programmable machine capable of carrying out complex actions automatically. Robots can be autonomous or controlled by an operator. As mobile robots and fixed robots, a broad category can be brought out in robotics, considering the ability to move the whole body from one point to another. Again, mobile robots can further consider the locomotion mechanisms used, which can be limbless, rolling, legged, or hybrid. From an application point of view, mobile robots have more benefits than fixed robots, which is a suitable reason for conducting sufficient research related to their design and control system.

Each type of locomotion mechanism offers some advantages which others might not, considering a specific application. In the case of a general-purpose application, where flexibility, dexterity, and stability are salient key parameters, legged robots, especially quadruped robots, provide more advantages. As Hybrid mobile robots combine basic locomotion, a combination of wheels and legs, is the best combination to answer the above requirements. Only a little research has been carried out from the literature about wheel-legged robots, especially quadruped wheel-legged robots. Quadruped wheel-legged robot design and development is the new trend in robot design due to the pool of possibilities they offer. This area needs to be considered for further research, as essential contributions can solve currently unsolved problems.

In the same track, this paper proposes a design for a quadruped wheel-legged robot, which explores a different combination of wheels, i.e., mecanum wheels with four legs. The design is provided considering different aspects, such as the mechanical and structural design of the robot's body, the electronic design, and the software design.

II. RELATED WORK

Many researchers proposed several designs for quadruped robots based on studies performed on quadruped animals. Each design is different from others considering specific parameters. More studies have been carried out on quadruped robots, while research on hybrid wheel-legged robots is ongoing. Let us discuss some of them (quadruped and wheel-legged robots), considering their designs and features.

Paper [1-3] deals with the MIT cheetah, a quadruped robot developed by the MIT (Massachusetts Institute of Technology) in 2013. , the accent was made on energy usage and saving in the locomotion mechanism using four design assumptions. The robot's power is in the range of 973 W with 0.5 cost of transport, which is comparable to that of running animals. A second version of the same robot was released in 2015 with higher goals in terms of performance. The robot was designed to reach 4.5 m/s on uneven terrains and jump over 400 mm obstacles at a speed of 2.5 m/s.

Boston Dynamics Corporation created a powerful quadruped robot named Big Dog [4]. The legs were designed with four DOF actuated with hydraulic actuators, as they can absorb and release energy during transition steps. The robot can exhibit a trotting gait at 0.8 m/s. The robot's motion is powered by a powerful 15HP gasoline engine and controlled by a Pentium processor with Quenix as Operating System. For environmental awareness, fifty sensors are incorporated to sense parameters like joint force and positions, ground contact, environment images, etc. The second version of big dog was introduced later, but the project was discontinued as the engine was noisy.

The same corporation introduced littleDog [5-6], a quadruped robot designed to navigate uneven and irregular terrains. The legs incorporate three joints, each driven by high gains servo motors. Tracking the motion for further studies used a total of twenty-two reflection balls fixed on the robot's body and joints.

Another robot introduced by Boston Dynamics Corporation in 2011 is the fastest quadruped robot ever developed, named. While being tethered, a speed of up to 45 km/h is achieved in laboratory conditions [7]. An advanced version was released in 2013 with the intention of creating the fastest untethered quadruped robot. This robot, powered by a methanol engine weighing 154 kg, with hydraulic actuators and a run speed of 32 km/h, was baptized Wildcat [8]. The robot exhibited good stability when tested outdoors with different gaits like trotting, galloping, bounding, etc.

Recently, Boston Dynamics released the most advanced quadruped robot, which is untethered and autonomous. This robot is named Spot and shares few design similarities with previous robots developed like BigDog and LS3 [9]. Spot can run at a speed of up to 5.76 km/h, weighing 30kg, with the ability to carry a load of 14 kg. Spot incorporates dedicated computers for overall control over its abilities. It can be remotely operated and achieve specific tasks autonomously. It features many sensors for surroundings awareness and electric actuators for motion. These specifications make Spot the most advanced light-footed, multi-terrain, quadruped robot.

A quadruped robot, very similar to Spot, is created by ETH Zurich for specific industrial and commercial operations involving challenging environments [10]. An example of such applications is rescue operations, search operations, working on a gas platform, and working at a commercial place. The robot is entitled ANYmal and weighs around 30kg, with the ability to carry an extra load other than the inbuilt equipment (batteries, sensors, microphones). ANYmal can execute different gaits like running, trotting, climbing, and others. The legs' motion is driven by electric torque-controllable actuators, which have the benefit of driving the joints at high precision [11].

A comparison between wheeled and legged robots is made in [12], considering different technical criteria such as maneuverability, transverse ability, controllability, terrain, land, efficiency, stability, cost-effectiveness, and navigation over obstacles. Both locomotion mechanisms complement each other. Exploring hybrid locomotion robots with wheels and legs is a good idea. ETH Zurich explores this track, an advanced version of the ANYmal robot, which features wheels on each leg for more mobility [13-16].

From the above literature, we can see that the trends in mobile robotics are moving towards hybrid robots, especially wheel-legged robots, as they offer more mobility and flexibility, which is very important for versatile robots designed to be used in different environments, and incredibly challenging environments. This paper will contribute to research in the same track by proposing a design for wheel-legged robots.

III. CONTRIBUTION

The design combines two types of locomotion mechanisms, namely: rolling locomotion and legged locomotion. A legged robot can be of either two, four, or multiple legs (most of the time, six legs maximum). A four-leg robot's configuration is generally referred to as a quadruped robot, offers good stability, and requires less energy than other leg configurations [17]. Another advantage is that a four legs configuration can also be used as a two-leg configuration (biped configuration) with an appropriate control system, making it the best configuration to adopt. Coming up to rolling locomotion, different wheels with different characteristics are available. Among those, omnidirectional wheels have more characteristics as they offer omnidirectionality without any steering. Hence, appending mecanum wheels to a quadruped robot might give rise to many possibilities as far as locomotion is concerned [18].

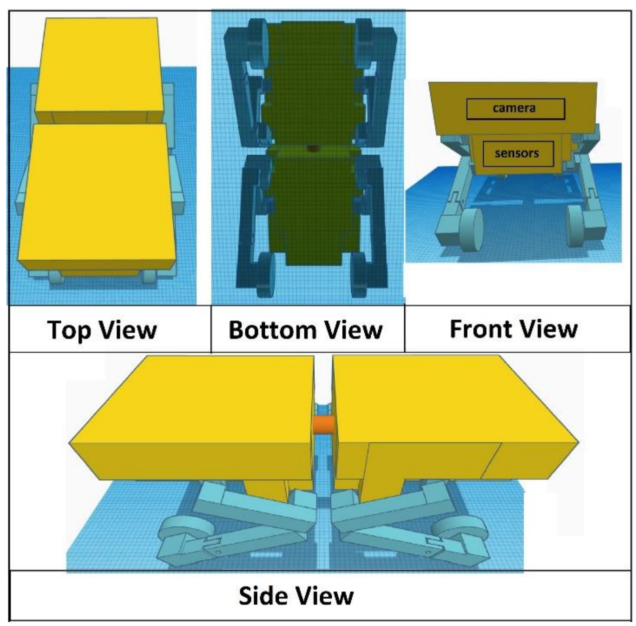

Figure 1: 3D Design of the robot (REVBOT)

The proposed design (figure 1) is discussed concerning three aspects: mechanical design, electronic design, and software design. The mechanical design deals with the design of wheeled legs, overall structural design, stabilization module, and other vital parts that constitute the proposed design's novelty. The electronic design discusses respect for the components required to drive the robot and achieve the goal. The last aspect of the design concerns the software architecture that enables researchers to interact with the robot and test their models through it.

A. Mechanical Design

The mechanical design is divided into four main parts: the overall structural design and layout, the design of the legs, the design of the spinal joint, and the design of the stabilization module. The proposed design is made with the following considerations:

- The robot's torso is divided into two parts, joined by a three DoF spinal axis
- Each part of the torso is divided into two levels. Each level houses specific components
- A leg has four DoF (hip, knee, and hock) and integrates a mecanum wheel with the ability to detect ground contact forces
- The robot has a flywheel-based gyroscopic stabilization module

a) Overall structural design and layout

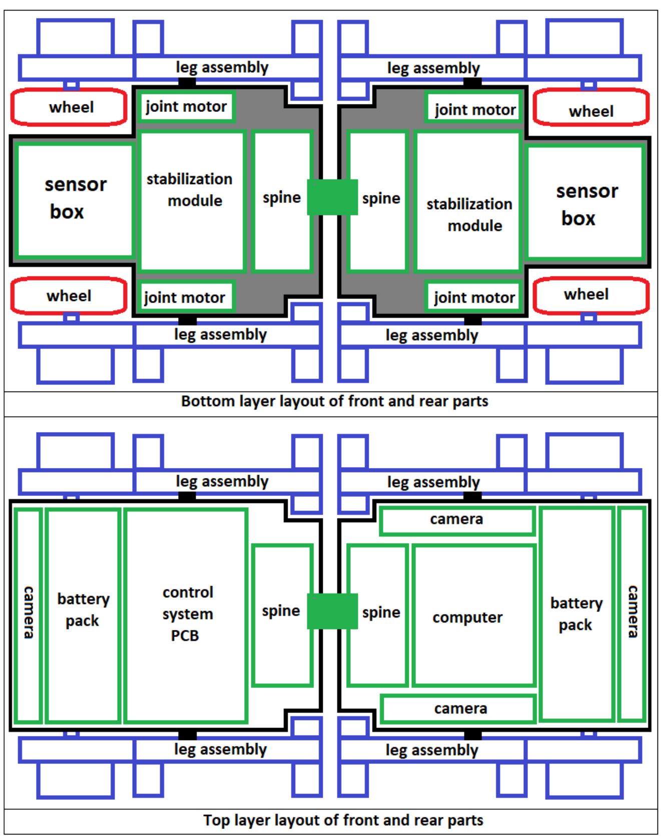

Figure 2: Structural design and components layout of the robot

To increase the versatility and flexibility of the robot, a three DoF (Degrees of Freedom) spinal joint is advisable. The whole body is divided into two parts based on that assumption. The structural design of the robot should consider the components that will be housed in terms of shape, space, and weight.

The robot's torso is divided into two parts linked by a spine. The structure of a single torso consists of 4 main parts, the front structure, the back structure, the chassis, and the top structure. The front part is divided into top and bottom sections. The top level is meant for the light source and vision camera, while the bottom level will house environmental sensors. The shape of the chassis considers the wheels and the lateral actuator of the hip joint. The chassis is divided into three sections, considering the components to be mounted. The back structure accommodates the spine mechanism, with a different pattern for both torso parts. The top structure is divided into two sections, with the lower one designed to enable the battery to be inserted and removed easily. The four parts assembled constitute half part of the torso.

Figure 2 shows the layout of each torso concerning the components to be accommodated. Each torso is divided into two levels, the top and bottom levels. The bottom or lower layer has three sections to accommodate a sensor box for terrain detection and analysis, a flywheel-based gyroscopic stabilization module, and the spine mechanism. Two actuators for the lateral swing of the hip joints are located between the first two sections. The upper layer or top level of the torso accommodates all electronic circuits necessary for the robot's working. It is divided into two main sections, the battery section and the circuit section. The whole-body structure can further be covered with the desired shape using aluminum sheets and other suitable materials.

b) Design of the legs

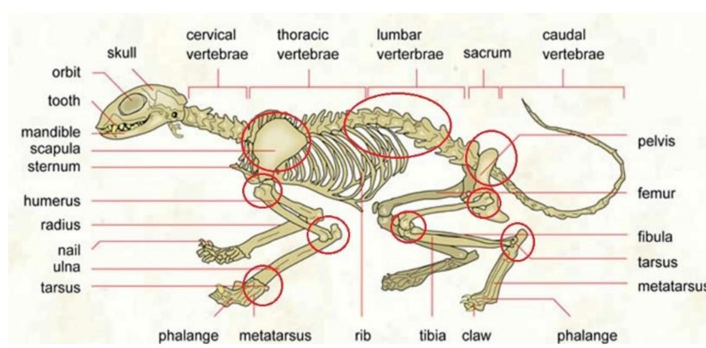

Figure 3: Skeleton of a cat highlighting different joints

Figure 3 shows the skeleton of a cat, highlighting the bones involved in its structure [19]. From the figure, the front and rear limbs have different structures. Let us consider the rear limbs as the inspiration for the design of the legs, as they are designed to carry more weight. We distinguish three prominent bones: the metatarsus, the tibia, and the femur.

Assuming each bone to be a link interconnected by joints, three joints are involved in the interconnection of these links: the ankle/wrist joint, which links the metatarsus and the tibia; the knee/elbow joint, which links the tibia and the femur, and finally the hip joint which links the pelvis and the femur. The same structure is reproduced with the help of 3D-printed parts and electric actuators, as shown in figure 4.

Few considerations are ensured while making the design; they are:

- The leg should be able to absorb shocks, which might occur during landing and disturbances
- The leg should be able to sense the force applied at the point of contact with the ground
- The leg should accommodate a wheel for hybrid locomotion

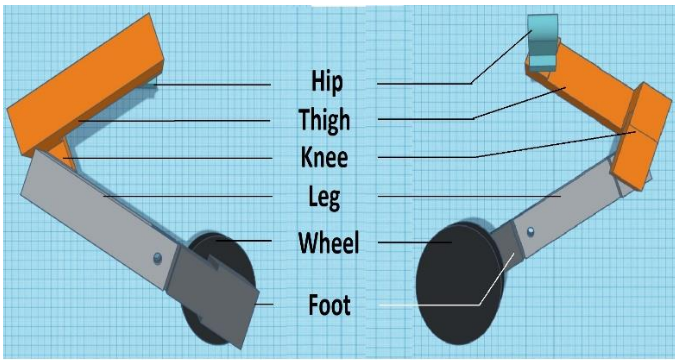

Figure 4: Design of the robot's limb

Following the abovementioned considerations, a dedicated shock absorber fulfills the shock absorption requirement. As the leg should sense forces and accommodate a wheel, a special arrangement is made to fulfill both requirements. Figure 5 shows the arrangement made to record forces through the wheels. Two force sensors are placed at 90 degrees on the wheel motor's side surfaces to record the applied horizontal and vertical forces. Two vibration dampers are placed between the force sensor and the walls of the enclosure to allow the wheel motor to move freely and transfer forces to the force sensors.

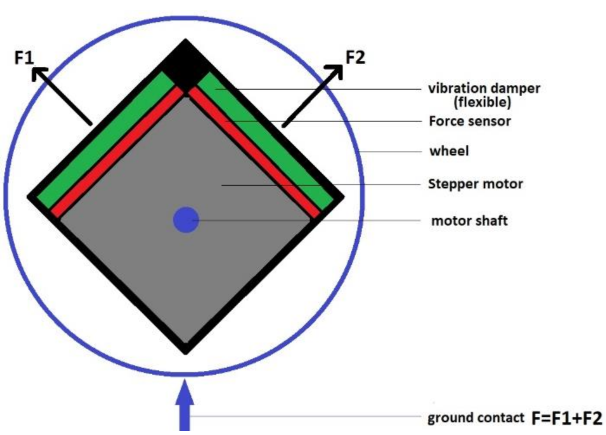

Figure 5: Structural diagram of the force detection mechanism

c) Design of the Spinal joint

A few quadruped robot designs include a spine. Depending on the design, spines can be either active or passive [20]. A single rotation axis can be used for the spine movement, either vertical or horizontal. Providing more flexibility implies generating complex movements, which can be more effective using multiple-axis spinal joints. The spinal movement plays a crucial role in maintaining the proposed design consisting of a three axes arrangement involving four geared DC motors. The motors are used to provide the swings required to mimic the behavior of an active spine. The arrangement is divided into two parts, which are combined using supports to form the spinal joint. Each part is accurately mounted onto each part of the torso. The spine assembly's first part on the front of the torso provides vertical and rotational swings. The second part of the spine assembly mounted on the back part of the torso provides a horizontal swing. Figure 6 shows a schematic of the spine assembly, in which 3D-printed parts are used as supports for the motors. The central support is designed to allow cables inside since the communication between both parts of the torso is wired. The robot's stability as it helps shift and adjusts the center of mass concerning the actual position of the robot's body. The design of a spinal joint requires some critical considerations:

- The number of rotation axes/degrees of freedom
- The weight to be beared
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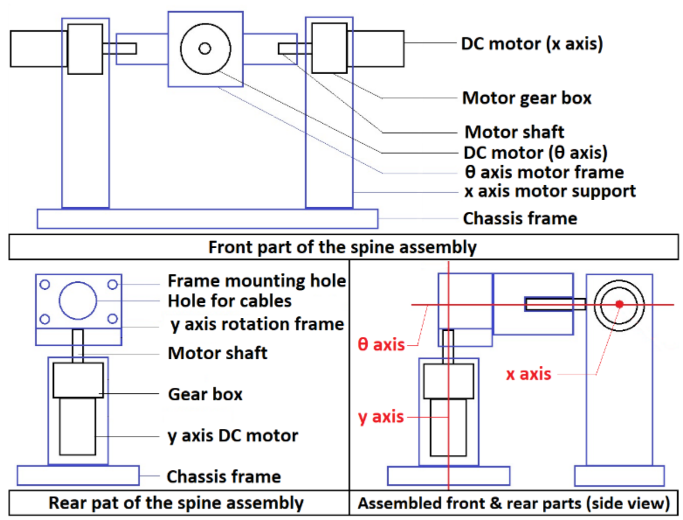

Figure 6: Structural diagram of the Spine Assembly

The proposed design consists of a three-axis arrangement involving four geared DC motors. The motors are used to provide the swings required to mimic the behavior of an active spine. The arrangement is divided into two parts, combined using supports to form the spinal joint. Each part is accurately mounted onto each part of the torso. The spine assembly's first part on the front of the torso provides vertical and rotational swings. The second part of the spine assembly mounted on the back part of the torso provides a horizontal swing. Figure 6 shows a schematic of the spine assembly, in which 3D-printed parts are used as supports for the motors. The central support is designed to allow cables inside since the communication between both parts of the torso is wired.

d) Design of the stabilization module

A *Gyroscopic stabilizer* can be defined as a control system that reduces the tilting movement of a body. It senses orientation and counteracts rotation by applying force to a large gyroscope [21]. On the other hand, a flywheel is known as a mechanical device that saves rotational energy using the principle of the conservation of angular momentum; this rotational energy is similar to kinetic energy that is proportional to the product of its moment of inertia and the square of its rotational speed [22]. Combining both principles with appropriate analysis helps generate the same force required to counter a disturbance. The flywheel-based gyroscopic stabilization module is the result of that combination. The design of the module is made following these considerations:

- The orientation of the wheel should be active
- The module should sense disturbances autonomously
- The module should be able to generate a counteracting force to maintain the robot's stability

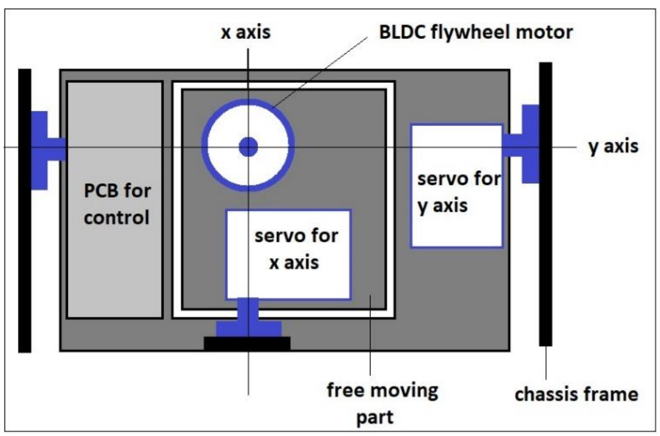

Figure 7: Structure diagram of the stabilization module

Depending on the situations the robot might come across, the choice to adopt either a passive or active adjustment is affected. In case of a huge number of disturbances, a passively actuated flywheel might not be efficient. To be able to maintain the stability under different types of disturbances, the flywheel rotation speed and axes should be controlled. Hence, our objective is to design a module which create generate a force in a specific direction as a response to a disturbance, to maintain the robot's stability. Figure 7 shows the structural design of the module, highlighting the components involved in the design and the rotation axes. The module design consists of a simple flywheel mounted on a BLDC motor, with two servo motors for position adjustment and a IMU sensor for disturbance sensing.

First and foremost, a specific position is defined as the stable position. If the actual values are different from the reference values corresponding to the stable position, such state is called as unstable; we should therefore perform some action to re-establish the robot's stability. This requires a force to be generated in a specific direction to cancel the effects of the unstable condition. That force can be generated with the flywheel based gyroscopic stabilization module through the control of flywheel rotation speed and the rotation angle on each axis through the servo motors. Finally, the stability can be cross-checked with the stability reference values through a feedback system. In case the robot doesn't experience any disturbance above a predefined threshold, the module is kept in sleeping mode, thereby saving energy.

B. Electronic Design

The electronic design consists of all the components required for the robot's work. The robot can be divided into modules to simplify the design and fabrication (figure 8). For instance, the coordination of all the modules is assured by a dedicated computer. The electronic design will be discussed concerning the following aspects: the main control unit, the vision unit, the communication unit, the sensing unit, the auxiliaries unit, the motion unit, and power unit.

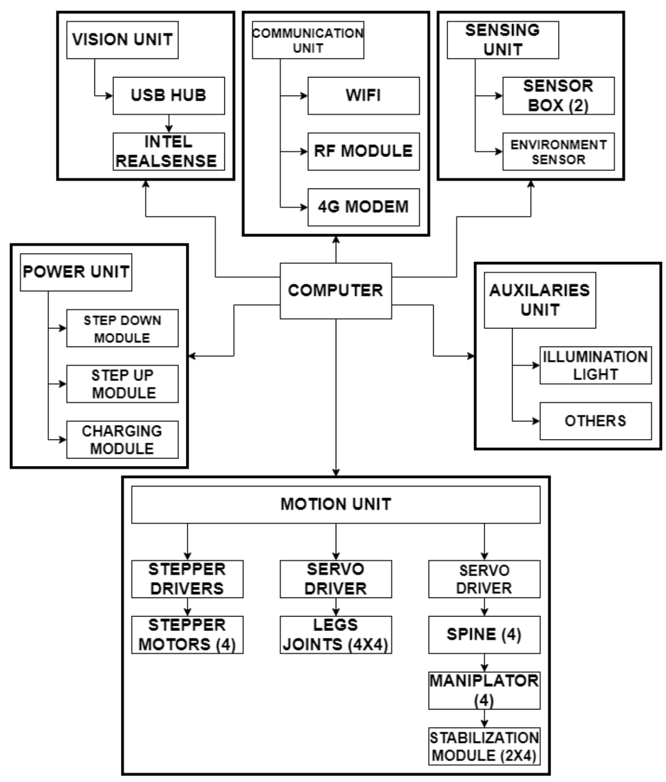

Figure 8: Electronic design architecture of the robot

a) Main Control Unit

Figure 9: Nvidia jetson nano embedded computer

The central control unit is just a computer board used to control the robot parts. As the robot has multiple parts consisting of sensors and actuators, coordination is required. Another critical aspect is the user interface which takes care of the communication between an operator. The user interface translates the commands sent by the operator and controls each part of the robot accordingly to execute the required task. Many single-board computers can be used for this purpose. However, a few considerations should be ensured; they are:

- The computer should have AI/ML capabilities
- The computer should be fast enough to process all the incoming data and respond accordingly
- The processor should be powerful enough
- The computer should have GPIOs pins to and support general-purpose sensors
- The computer should be of a small form factor and should have less power consumption and power rating

An example of such a computer can be found in the Nvidia Jetson series of embedded computers. Nvidia Jetson series embedded computers are engineered for autonomous machines design working with AI. A good choice for the current application is the Nvidia Jetson Nano (figure 9), which is a small, powerful computer designed to power entry-level edge AI applications and devices.

b) Vision Unit

Figure 10: intel real sense depth camera

The ability to move requires a locomotion mechanism and some ways of understanding the environment through sensors [23]. The best way to be aware of its environment is through vision. Hence, high-end cameras can be used to enable the robot's vision. Detecting obstacles and adapting the locomotion requires input, which can be taken from a camera. More insights are possible through depth cameras (figure 10), which provide a rendered 3D image of the environment. The vision unit consists of four cameras to enable a 360-degree view of the surroundings.

c) Communication Unit

The communication unit is meant for the robot to communicate with an external operator. The robot can be either teleoperated or autonomous. Even though the robot is autonomous, a task should be assigned to fulfill it. As the robot is designed as a research platform on wheel-legged locomotion, an autonomous task could be to reach a specific target point with suitable locomotion. Irrespective of the type of control, it is mandatory to communicate with the robot. Communication is achieved in our case by employing three modules: a Wi-Fi module, an RF module, and a 4G USB modem. The Wi-Fi module helps to establish communication between the robot and the operator from an existing Wi-Fi network. The same can be used as a hotspot, creating a personal network on which the operator can connect and control the robot. The same goal is achievable through an RF module connected to a remote readily available. The last module used for communication is the 4G modem, which supplies uninterrupted internet access to the robot. Thereby allowing a distant connection between the robot and the operator's system. This means allows the robot to be operated almost everywhere, as long as an internet connection is available.

d) Sensing Unit

The sensing unit consists of two main modules: the sensor box and the environment sensing unit. The sensor box is meant to detect, identify, and analyze the terrain on which the robot is being operated. On the other hand, the environment sensing unit features additional sensors for gathering more data from the outside world.

The sensor box for terrain analysis consists of different sensors which collect data and use some algorithms to execute different tasks [24]. The sensor box features a camera, some laser distance sensors, an accelerometer and a gyroscope, and temperature and humidity. Each of these sensors is integrated to achieve different tasks. The sensor box will get the robot's height or the clearance from the ground using laser distance sensors; the whole combination is mounted on a servo motor for adjustment. The intention while combining these sensors is to collect valuable data about the terrain to be navigated (clearance of the robot from the ground, path tracking, and terrain perception) with the help of parameters like speed, orientation, distances, and images.

The environment sensing unit is a two-way communication module that implies an interaction between the robot and its environment. A few parameters like humidity, temperature, atmospheric pressure, vibrations, sound, and external forces need to be monitored for the robot to successfully and accurately execute a task. Monitoring such parameters requires temperature and humidity sensors, atmospheric pressure sensors, force sensors, and microphones for sound sensing. All these sensors are assembled into a single unit, which is the environment sensing unit.

e) Auxilaries Unit

The auxiliaries unit deals with significant but not mandatory units, like GPS modules and lighting. This unit controls the external module that might be connected to the robot, such as LIDAR, External sensors, thermal cameras, etc. GPS, lighting, indicators, and others modules. This unit has its importance as the design of the robot stands on a modular concept, where different modules can be added to the existing design.

f) Motion Unit

The motion unit controls the robot's motion in all aspects. All moving parts are driven through electrical actuators. The electrical motors used as actuators for the legs joints, spinal joint assembly, stabilization module, module adjustment, and wheels need to be driven consequently [25]. The wheels are driven with a stepper motor to control each step and serve for wheel odometry. BLDC geared motors drive the leg joints and geared DC motors do the spinal joint assembly. The stabilization module and several other modules, like the camera and the sensor box, use servo motors. Controlling such drives requires appropriate drivers, shown in the electronic design architecture (figure 8). The motion unit features stepper drivers for stepper motors, BLDC ESCs for BLDC motors, DC motors ESCs for DC motors, and PWM servo control modules for the ESCs and servo motors.

g) Power Unit

A mobile robot should only depend on a fixed power source as it will allow the robot to move freely. The best way to provide mobility freedom to mobile robots is by using a power source that can be carried onboard and recharged whenever needed. This problem can be addressed by rechargeable LiPo batteries, which can be assembled to meet the power needs of the robot and recharged when they get discharged. As the robot consists of many units with different requirements, thereby different power needs, the power source should be able to fulfill those needs and deliver the exact power needs of each unit. The power unit consists of buck and boost converters which set the same power required by each unit in the system.

C. Software Design

The platform being developed should serve for research on a framework for an autonomously driven robot, mobility of quadruped robots, stability of quadruped robots, and so on. Therefore, the platform should have all the components of such a system. The electronic design architecture considers such parameters by integrating some essential components grouped in different units. The units should be accessible through a framework to allow the user to control the robot as he wants.

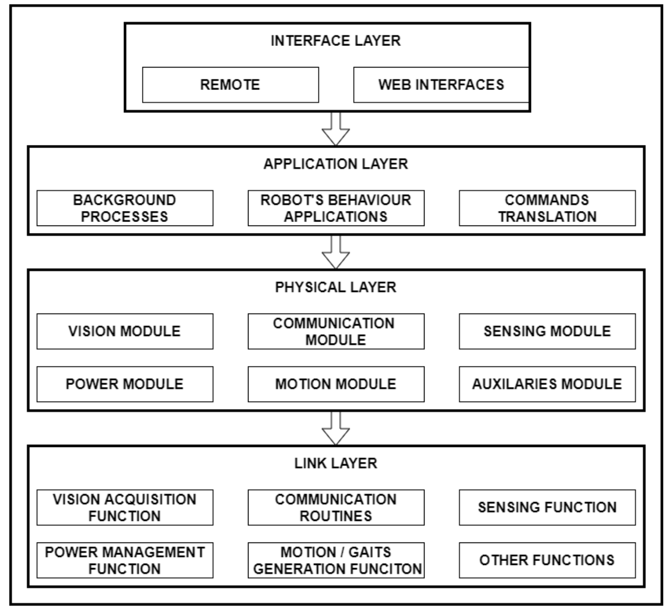

Figure 10: four layers software architecture

Figure 10 shows the proposed software architecture, a four-layer model consisting of a physical layer, a link layer, an application layer, and an interface layer. The physical layer represents all the physical components that constitute the robot per the electronic design. The physical layer acts on the physical components using six units: the vision unit, the communication unit, the sensing unit, the power unit, the motion unit, and finally, the auxiliaries unit. The link layer links the application layer to the physical layer using dedicated functions or algorithms. The link layer uses the resources available in the physical layer to execute tasks dictated by the application layer. The functions used in the link layer are categorized into six groups the power management function, the vision acquisition function, the communication function, the motion generation function, the stability function, and other functions that might arise in later stages. A mapping function can be grouped in the vision acquisition function and use the vision module to establish a map if the application layer requests.

Similarly, the communication function will use the communication module to send and receive data. Retrieving data from all sensors present in the physical layer is possible through the sensing function. For example, a request to move in a specific direction will be processed through the motion generation function in the link layer and executed by the motion module in the physical layer. The application layer consists of three main groups: the background processes, the robot's behavior applications, and the command acquisition. Any command coming from the upper layer is first processed by the command acquisition function, which communicates with the functions defining the robot's behavior. As the robot's behavior depends on many parameters, few functions are kept as background processes to retrieve required data. The last layer is the interface layer, which allows any user to interact with the robot through commands. The interface can be of any form, like a web application, a mobile application, or a remote with physical buttons.

IV. Conclusion

Mobile robots have many applications and can substitute humans in performing specific tasks. Substituting humans in accomplishing these tasks implies working in different environments that can sometimes be difficult to tame due to irregularities and obstacles. Hence, robots' locomotion system is essential and must be elaborated carefully. This paper proposed a design for wheel-legged quadruped robots with omnidirectional wheels and an active spine to bring more flexibility to the robot's movement. The design is discussed considering different aspects, and an outlook of the robot is presented. The robot is designed as a research platform for quadrupedal locomotion research, as the trend is toward wheel-legged robots because of the possibilities they offer in terms of mobility. In future work, we intend to develop the robot concerning the proposed design and demonstrate the ability to move according to some implemented motions' behaviors.

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